Byung-Kwan Lee

Ph.D., School of Electrical Engineering at KAIST

Research Scientist at NVIDIA

bklee.nano@gmail.com · byungkwanl@nvidia.com · Homepage

LinkedIn · Github · Google Scholar · Huggingface

Research Interest

Building **efficient, high-performing Vision-Language Models (VLMs)**, with focus on:

- Distillation
- RL
- Agent
- Pruning

Work Experience

NVIDIA • ***Research Scientist*** *Oct. 2025 - Current*

LEAD **ZPPO: Zone of Proximal Policy Optimization** [COMPLETED] *RL post-training that keeps the teacher inside the prompt rather than the policy gradient, using BCQ/NCQ question reformulations and a prompt replay buffer to lift small students on hard questions without policy drift.*

Under Review **U.S. Provisional Patent Filed 2026** **ArXiv Release** **Internal Release**

LEAD **AXPO: Agent eXplorative Policy Optimization** [COMPLETED] *Agentic RL training that recovers tool usage through tool-call resampling, improving multimodal reasoning performance against larger baselines.*

Under Review **U.S. Provisional Patent Filed 2026** **ArXiv Release** **Internal Release** **Internal Tech Transfer**

LEAD **Masking Teacher and Reinforcing Student** [COMPLETED] *Mask-progressive RL distillation that gradually unmasks teacher weights and uses offline RL with accuracy & distillation rewards.*

CVPR 2026 Accept **U.S. Non-Provisional Patent Filed 2026** **ArXiv Release** **Internal Release**

LEAD **Unified RL & Imitation Learning for VLMs** [COMPLETED] *Combines RL with adversarial imitation, using an LLM-based discriminator and multi-teacher guidance to build lightweight yet powerful VLMs.*

NeurIPS 2025 Accept **U.S. Patent Upgraded to Non-Provisional 2025** **ArXiv Release** **Internal Release**

LEAD **GenRecal** [COMPLETED] *Cross-architecture VLM distillation via a Recalibrator that aligns heterogeneous token representations regardless of vocabulary, token splits, or index ordering.*

ECCV 2026 Accept **U.S. Patent Upgraded to Non-Provisional 2025** **Internal Tech Transfer**

LEAD **VLsI: Verbalized Layers-to-Interactions** [COMPLETED] *Hardened the verbalizer-based layer-wise distillation into a production-ready recipe and led its internal tech transfer into NVIDIA's VLM development pipeline.*

CVPR 2025 Accept **U.S. Patent Upgraded to Non-Provisional 2025** **ArXiv Release** **Internal Release** **Internal Tech Transfer**

NVIDIA • ***Research Intern*** *Oct. 2024 - Oct. 2025*

LEAD **Unified RL & Imitation Learning for VLMs** [INITIATED] *First formulation of the RL + adversarial imitation training pipeline; initial experiments and team setup.*

U.S. Provisional Patent Filed 2025

LEAD **GenRecal** [INITIATED] *Initial design of the Recalibrator framework for cross-tokenizer VLM distillation; first proof-of-concept and internal demo.*

U.S. Provisional Patent Filed 2025 **ArXiv Release** **Internal Release**

LEAD **VLsI: Verbalized Layers-to-Interactions** [INITIATED] *Layer-wise distillation using intermediate verbalizers, enabling small VLMs (2B/7B) to align with large VLMs' reasoning progression and outperform GPT-4V.*

CVPR 2025 Accept **U.S. Provisional Patent Filed 2025** **ArXiv Release** **Internal Release**

Education

KAIST *Mar. 2020 - Aug. 2025*
Ph.D., School of Electrical Engineering
◦ GPA: 3.77/4.3

◦ Dissertation: *Building High-performing, Efficient-size Vision Language Models: Merge, Modify, and Distill* [Link]

KAIST *Mar. 2018 - Feb. 2020*
M.S., The Cho Chun Sik Graduate School of Green Transportation
◦ GPA: 3.72/4.3

◦ Thesis: *Training Encoder-Attention through Fully-Connected CRFs for Efficient End-to-End Lane Detection Model* [Link]

Hanyang University *Mar. 2014 - Feb. 2018*
B.S., Mathematics and Electronic Engineering
◦ GPA: 3.86/4.5

◦ Thesis: *Learning Rank of Evaluated Songs for Classifying Unknown Music by Convolutional Neural Network* [Link]

Publications

Overall	18 Accepts · 4 Pending · 2 Tech Reports
CV	5 CVPR · 2 ICCV · 3 ECCV · 1 ICIP
ML	3 NeurIPS · 1 ICLR
NLP	1 ACL · 1 EMNLP
Journal	1 Pattern Recognition

▪ **79%** (19/24) First & Lead Author
▪ **21%** (5/24) Co-Author

1. **Zone of Proximal Policy Optimization: Teacher in Prompts, Not Gradients**
   Byung-Kwan Lee, Ximing Lu, Shizhe Diao, Minki Kang, Saurav Muralidharan, Karan Sapra, Andrew Tao, Pavlo Molchanov, Yejin Choi, Yu-Chiang Frank Wang, Ryo Hachiuma

   Under Review [Paper][Project]

   U.S. Patent Application Filed (Provisional), NVIDIA Research, 2026

2. **SpatialClaw: Rethinking Action Interface for Agentic Spatial Reasoning**
   Seokju Cho, Ryo Hachiuma, Abhishek Badki, Hang Su, Byung-Kwan Lee, Chan Hee Song, Sifei Liu, Subhashree Radhakrishnan, Seungryong Kim, Yu-Chiang Frank Wang, Min-Hung Chen

   Under Review [Paper][Project][Code]

   U.S. Patent Application Filed (Provisional), NVIDIA Research, 2026

3. **Agent eXplorative Policy Optimization for Multimodal Agentic Reasoning**
   Minki Kang, Shizhe Diao, Ryo Hachiuma, Sung Ju Hwang, Pavlo Molchanov, Yu-Chiang Frank Wang, Byung-Kwan Lee

   Under Review [Paper][Project]

   U.S. Patent Application Filed (Provisional), NVIDIA Research, 2026

4. **Hide to See: Reasoning-prefix Masking for Visual-anchored Thinking in VLM Distillation**
   Seonghoon Yu, Dongjun Nam, Byung-Kwan Lee†, Jeany Son†

   Under Review [Paper][Code]

5. **Why and When Visual Token Pruning Fails? A Study on Relevant Visual Information Shift in MLLMs Decoding**
   Jiwan Kim, Kibum Kim, Wonjoong Kim, Byung-Kwan Lee, Chanyoung Park

   European Conference on Computer Vision (ECCV), 2026 [Paper][Project]

6. **GenRecal: Generation after Recalibration from Large to Small Vision Language Models**
   Byung-Kwan Lee, Ryo Hachiuma, Yong Man Ro, Yu-Chiang Frank Wang, and Yueh-Hua Wu

   European Conference on Computer Vision (ECCV), 2026 [Paper][Project]

   U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2025

7. **Masking Teacher and Reinforcing Student for Distilling Vision-Language Models**
Byung-Kwan Lee, Yu-Chiang Frank Wang, Ryo Hachiuma
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 [Paper]
U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2026

8. **Recursive Think-Answer Process for LLMs and VLMs**
Byung-Kwan Lee*, Youngchae Chee*, Yong Man Ro
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings, 2026 [Paper][Project]

9. **RefineBench: Evaluating Refinement Capability in Language Models**
Young-Jun Lee*, Seungone Kim*, Byung-Kwan Lee, Minkyeong Moon, Yechan Hwang, Jong Myoung Kim, Graham Neubig, Sean Welleck, Ho-Jin Choi
International Conference on Learning Representations (ICLR), 2026 [Paper][Project]
Best Runner-Up Award, Workshop for Multi-Turn Interactions in Large Language Models, NeurIPS 2025 (Oral, Top 1%) [Link]

10. **Unified Reinforcement and Imitation Learning for Vision-Language Models**
Byung-Kwan Lee, Ryo Hachiuma, Yong Man Ro, Yu-Chiang Frank Wang, Yueh-Hua Wu
Neural Information Processing Systems (NeurIPS), 2025 [Paper][Project]
U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2025

11. **MultiVerse: A Multi-Turn Conversation Benchmark for Evaluating Large Vision and Language Models**
Young-Jun Lee, Byung-Kwan Lee, Jianshu Zhang, Yechan Hwang, Byungsoo Ko, Han-Gyu Kim, Dongyu Yao, Xuankun Rong, Eojin Joo, Seung-Ho Han, Bowon Ko, Ho-Jin Choi
IEEE/CVF International Conference on Computer Vision (ICCV), 2025 [Paper][Project]
Workshop for Knowledge-Intensive Multimodal Reasoning, ICCV 2025 [Link]

12. **VLsI: Verbalized Layers-to-Interactions from Large to Small Vision Language Models**
Byung-Kwan Lee, Ryo Hachiuma, Yu-Chiang Frank Wang, Yong Man Ro†, Yueh-Hua Wu†
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025 [Paper][Project]
U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2025

13. **Phantom of Latent for Large Language and Vision Models**
Byung-Kwan Lee, Sangyun Chung, Chae Won Kim, Beomchan Park, Yong Man Ro
Technical Report [Paper][Code][Huggingface Model]

14. **TroL: Traversal of Layers for Large Language and Vision Models**
Byung-Kwan Lee, Sangyun Chung, Chae Won Kim, Beomchan Park, Yong Man Ro
Empirical Methods in Natural Language Processing (EMNLP), 2024 [Paper][Code][Huggingface Model]

15. **Meteor: Mamba-based Traversal of Rationale for Large Language and Vision Models**
Byung-Kwan Lee, Chae Won Kim, Beomchan Park, Yong Man Ro
Neural Information Processing Systems (NeurIPS), 2024 [Paper][Code][Huggingface Model]
31st Samsung HumanTech Paper Awards in section of Computer Science & Engineering

16. **MoAI: Mixture of All Intelligence for Large Language and Vision Models**
Byung-Kwan Lee, Beomchan Park, Chae Won Kim, Yong Man Ro
European Conference on Computer Vision (ECCV), 2024, [Paper][Code][Huggingface Model]

17. **CoLLaVO: Crayon Large Language and Vision mOdel**
Byung-Kwan Lee, Beomchan Park, Chae Won Kim, Yong Man Ro
Findings of the Association for Computational Linguistics (ACL), 2024, [Paper][Code][Huggingface Model]
2024 KCC XAI Workshop, Best Paper Awards

18. **Causal Unsupervised Semantic Segmentation**
Junho Kim*, Byung-Kwan Lee*, and Yong Man Ro
Journal of Pattern Recognition [Paper][Code]

19. **Mitigating Adversarial Vulnerability through Causal Parameter Estimation by Adversarial Double Machine Learning**
Byung-Kwan Lee*, Junho Kim*, and Yong Man Ro
IEEE/CVF International Conference on Computer Vision (ICCV), 2023 [Paper][Code]

20. **Mitigating Dataset Bias in Image Captioning through CLIP Confounder-free Captioning Network**
YeonJu Kim, Junho Kim, Byung-Kwan Lee, Sebin Shin, and Yong Man Ro
IEEE International Conference on Image Processing (ICIP), 2023 [Paper][Code]

21. **Demystifying Causal Features on Adversarial Examples and Causal Inoculation for Robust Network by Adversarial Instrumental Variable Regression**
Junho Kim*, Byung-Kwan Lee*, and Yong Man Ro
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023 [Paper][Code]

22. **Masking Adversarial Damage: Finding Adversarial Saliency for Robust and Sparse Network**
Byung-Kwan Lee*, Junho Kim*, and Yong Man Ro
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022 [Paper][Code]

23. **Distilling Robust and Non-Robust Features in Adversarial Examples by Information Bottleneck**
Junho Kim*, Byung-Kwan Lee*, and Yong Man Ro
Neural Information Processing Systems (NeurIPS), 2021 [Paper][Code]

24. **Towards Adversarial Robustness of Bayesian Neural Network through Hierarchical Variational Inference**
Byung-Kwan Lee*, Youngjun Yu, and Yong Man Ro
Technical Report [Paper][Code]

Reviewer Experience

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)	*Journal*
IEEE Transactions on Neural Networks and Learning Systems (TNNLS)	*Journal*
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)	*Journal*
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)	*Conference*
IEEE/CVF International Conference on Computer Vision (ICCV)	*Conference*
European Conference on Computer Vision (ECCV)	*Conference*
International Conference on Learning Representations (ICLR)	*Conference*
Neural Information Processing Systems (NeurIPS)	*Conference*
International Conference on Machine Learning (ICML)	*Conference*
AAAI Conference on Artificial Intelligence (AAAI)	*Conference*
Association for Computational Linguistic (ACL)	*Conference*
Empirical Methods in Natural Language Processing (EMNLP)	*Conference*

Invited Talks and Awards

Silver Reviewer Award, International Conference on Machine Learning (ICML)	*2026*
Best Runner-Up Award (Oral, Top 1%), Multi-Turn Interactions in LLMs Workshop @ NeurIPS	*2025*
Invited Talk in Kongju University	*2025*
Invited Talk in Hanyang University ERICA	*2025*
Invited Talk in Kookmin University	*2025*
Invited Talk in Dongseo University	*2025*

31st Samsung HumanTech Paper Awards in section of Computer Science & Engineering	*2025*
KCC XAI Workshop, Best Paper Awards	*2024*
Invited Talk in Korea Institute of Science and Technology Information (KISTI)	*2024*
Invited Talk in NAVER HyperCLOVA X	*2024*
Invited Talk in Kakao Brain	*2024*
1st Award for Research Presentation in Center for Applied Research in Artificial Intelligence (CARAI)	*2023*
GIRE Research Mentor, Seoul International School – "MUsE", Journal of Student Research [Publication]	*2023*
KAIST SW IT Academy, AI Project Director of 1st Award Topic	*2023*
Invitation to Presenter in Korean Conference on Computer Vision (KCCV)	*2023*
KAIST SW IT Academy, Lecturer of Data Structure & Algorithm, Deep Learning, Computer Vision	*2023*
Invitation to Presenter in Korean Conference on Computer Vision (KCCV)	*2022*
2nd Award for Research Presentation in Center for Applied Research in Artificial Intelligence (CARAI)	*2022*
KAIST Fellowship	*Mar. 2020 - Feb. 2024*
National Government Fellowship	*Mar. 2018 - Feb. 2020*
1st President's Award for International Autonomous Vehicle Competition, Team Leader [YouTube][Talk]	*2018*